

All-Clock Rigidity for Centered Three-Window Möbius Capacity

RH research program

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Abstract

We introduce a centered, deliberately noncausal local-table model for the distance-two Möbius capacity problem. At phase r , a sign table reads $\mu_0(n-1), \mu(n), \mu(n+1)$, and universal safety forbids two $+1$ outputs two steps apart for every ternary input word. The complete three-shift terminal-log table law reduces every fixed-clock limit to nonnegative exact-support densities.

Positive projection turns the 2^{27} local tables into 512 relations. Relation saturation then gives an exact tropical trace on the eight subsets of $\{-1, 0, +1\}$ for every clock. For $q \geq 3$ a multi-affine argument reduces an optimizer to four antipodally symmetric states; the self-loop clock $q = 2$ is a genuine exception and needs a one-sign state. Writing $K_j = \prod_p (1 - j/p^2)$, we obtain

$$C(1) = K_2 - K_3, \quad C(2) = \frac{3K_2 - K_3}{4}, \quad C(3) = \frac{3K_1}{8}, \quad C(4) = \frac{2K_1}{3},$$

and $C(6) = K_1/8 + K_2/2$, strictly above the corresponding one-site value $3K_1/8$.

Despite these finite-clock memory gains, the all-clock endpoint is rigid. A phasewise marginal identity on square-support clocks charges every adjacent pair by one common center weight, recovering exactly the one-site path-MWIS value. Divisibility lifting then proves

$$\sup_{q < \infty} C(q) = B_\infty, \quad C(q) < B_\infty \quad \text{for every finite } q.$$

The maximum is formed only after each fixed-table terminal limit. No growing clock, ordinary Cesàro, causal-transducer, adaptive-capacity, or higher even-correlation conclusion is asserted.

Keywords: Möbius function; terminal logarithmic average; centered local rule; max-plus optimization; squarefree density; nonattained supremum.

1 Model, densities, and principal theorem

Write $\mu_0(t) = \mu(t)$ for integers $t \geq 1$, and $\mu_0(t) = 0$ for $t \leq 0$. A *terminal clock* is any function

$$1 \leq \omega(X) \leq X, \quad \omega(X) \longrightarrow \infty. \quad (1)$$

Fix a positive integer q . A centered q -phase table family is

$$F_r : \{-1, 0, +1\}^3 \longrightarrow \{-1, +1\}, \quad r \in \mathbb{Z}/q\mathbb{Z}.$$

Its output at n reads one future coordinate:

$$\varepsilon_n = F_{n \bmod q}(\mu_0(n-1), \mu(n), \mu(n+1)). \quad (2)$$

Thus centered is a new noncausal data type, not an online transducer. The family is *universally distance-two safe* when

$$\neg(F_r(a, b, c) = +1 \text{ and } F_{r+2}(c, d, e) = +1) \quad (3)$$

for every r and every $a, b, c, d, e \in \{-1, 0, +1\}$. The shared letter c is the only coupling between the two windows.

For fixed q, F, ω , put

$$L_{q,X}(F) = \frac{1}{\log \omega(X)} \sum_{X/\omega(X) < n \leq X} \frac{\mu(n) F_{n \bmod q}(\mu_0(n-1), \mu(n), \mu(n+1))}{n}. \quad (4)$$

The complete three-shift law recalled below proves that the limit $L_q(F) = \lim_{X \rightarrow \infty} L_{q,X}(F)$ exists for every clock (1). Only after these fixed-table limits are formed do we set

$$C(q) = \max_{\substack{F \text{ fixed and} \\ \text{universally safe}}} |L_q(F)|. \quad (5)$$

Let $\mathcal{I} = \{L, C, R\}$ with shifts

$$a_L = +1, \quad a_C = 0, \quad a_R = -1, \quad (6)$$

so that $\mu_0(n - a_L), \mu_0(n - a_C), \mu_0(n - a_R)$ are the entries in (2). For $S \subseteq \mathcal{I}$, set

$$B_{p,S} = \{a_i \bmod p^2 : i \in S\}, \quad \nu_{p,S} = |B_{p,S}|,$$

where the set is deduplicated modulo p^2 , and

$$\tau_{p,S}(r) = \#\{b \in B_{p,S} : b \bmod p = r \bmod p\}.$$

The squarefree phase density is

$$\Theta_{q,r}(S) = \frac{1}{q} \prod_{p \nmid q} \left(1 - \frac{\nu_{p,S}}{p^2}\right) \prod_{p \parallel q} \left(1 - \frac{\tau_{p,S}(r)}{p}\right) \prod_{p^2 \mid q} \mathbf{1}_{\{r \bmod p^2 \notin B_{p,S}\}}. \quad (7)$$

In particular $\Theta_{q,r}(\emptyset) = 1/q$, and

$$\sum_{r \bmod q} \Theta_{q,r}(S) = K_{|S|}, \quad K_j = \prod_p \left(1 - \frac{j}{p^2}\right), \quad K_0 = 1. \quad (8)$$

The exact-support density is

$$\Pi_{q,r}(U) = \sum_{W \subseteq \mathcal{I} \setminus U} (-1)^{|W|} \Theta_{q,r}(U \cup W). \quad (9)$$

It is nonnegative and $\sum_{U \subseteq \mathcal{I}} \Pi_{q,r}(U) = 1/q$.

Theorem 1.1 (Centered capacity and all-clock rigidity). *For every fixed q , every safe table family has the terminal limit (4) for every clock (1). For $x, y \in \{-1, 0, +1\}$, let $S(x, y) \subseteq \{L, C, R\}$ contain C , contain L exactly when $x \neq 0$, and contain R exactly when $y \neq 0$. Define*

$$\lambda_r(x, y) = 2^{-\mathbf{1}_{\{x \neq 0\}} - \mathbf{1}_{\{y \neq 0\}}} \Pi_{q,r}(S(x, y)) \quad (10)$$

for $x, y \in \{-1, 0, +1\}$, and, for $U, V \subseteq \{-1, 0, +1\}$,

$$\mathcal{K}_r(U, V) = \sum_{\substack{x \notin U \\ y \in V}} \lambda_r(x, y). \quad (11)$$

Then the exact all- q capacity is

$$C(q) = \sum_{\gamma} \max_{\substack{Y_r \subseteq \{-1, 0, +1\} \\ r \in \gamma \text{ cyclic}}} \sum_{r \in \gamma} \mathcal{K}_r(Y_{r-2}, Y_r), \quad (12)$$

where γ runs over the $\gcd(q, 2)$ cycles of $r \mapsto r + 2$. Equivalently, each summand is the tropical trace of a cyclic product of exact 8×8 transition matrices.

For $q \geq 3$, some optimizer in (12) uses only the four antipodally symmetric subsets. With

$$u(Y) = \left(\mathbf{1}_{\{0 \in Y\}}, \frac{|Y \cap \{-1, +1\}|}{2} \right) \in \{0, 1\}^2$$

and

$$(a_r, b_r, c_r, d_r) = (\Pi_{q,r}(\{C\}), \Pi_{q,r}(\{L, C\}), \Pi_{q,r}(\{C, R\}), \Pi_{q,r}(\{L, C, R\})),$$

the compressed transition is

$$a_r(1 - u_0)v_0 + b_r(1 - u_1)v_0 + c_r(1 - u_0)v_1 + d_r(1 - u_1)v_1. \quad (13)$$

This compression is not a proof for $q = 1, 2$; their self-loops retain all eight states. The first exact values are

$$C(1) = K_2 - K_3, \quad C(2) = \frac{3K_2 - K_3}{4}, \quad (14)$$

$$C(3) = \frac{3K_1}{8} = \frac{9}{4\pi^2}, \quad C(4) = \frac{2K_1}{3} = \frac{4}{\pi^2}, \quad (15)$$

$$C(6) = \frac{K_1}{8} + \frac{K_2}{2}. \quad (16)$$

At $q = 6$, this strictly exceeds the embedded one-site value $3K_1/8$.

Let B_∞ be the RH-375 one-site square-clock endpoint, recalled in (27) below. Then

$$\boxed{\sup_{q < \infty} C(q) = B_\infty, \quad C(q) < B_\infty \quad (q < \infty).} \quad (17)$$

Both signs of every nonzero optimum are attained.

2 The analytic reduction to nonnegative relations

RH-394, Theorem 1.1, equation (8), Theorem 1.2, equations (11)–(12), and Corollary 1.3 (printed/PDF pages 2–3) prove the complete fixed three-shift table law for every terminal clock [2]. Applied to

$$H_r(x, z, y) = zF_r(x, z, y),$$

that law says that the limit in (4) is the sum, over exact support strata U , of $\Pi_{q,r}(U)$ times the uniform sign average of H_r on that stratum. Equations (7)–(9) are its phase-resolved densities. Its proof inherits the odd-correlation input of Tao–Teräväinen and the two-point input of Tao [3, 4]; we do not re-invoke those remote theorems with new quantifiers.

Lemma 2.1 (Positive projection). *From a safe family F , change every $+1$ output at a cell with center $z \neq +1$ to -1 , leaving all other cells unchanged. The resulting family F^+ is safe and*

$$L_q(F^+) \geq L_q(F).$$

Put

$$A_r = \{(x, y) \in \{-1, 0, +1\}^2 : F_r^+(x, +1, y) = +1\}. \quad (18)$$

Then

$$L_q(F^+) = \sum_{r \bmod q} \sum_{(x, y) \in A_r} \lambda_r(x, y). \quad (19)$$

Every one of the 512 relations occurs, with 2^{18} full sign-table preimages before projection.

Proof. If $z = -1$, changing the output from $+1$ to -1 changes the score zF from -1 to $+1$; if $z = 0$, the score is unchanged. Deleting $+1$ outputs cannot create a forbidden pair, proving the first claim pointwise before taking any limit.

The all-minus table contributes $H = -z$, whose sign average is zero on every stratum with nonzero center and which vanishes when the center is zero. Turning a center- $+1$ cell $(x, +1, y)$ from minus to plus changes the score by two. On its exact support stratum, the specified nonzero signs have density $2^{-|U|}\Pi_{q,r}(U)$. Multiplication by two leaves exactly (10), proving (19). The nine center- $+1$ cells determine A_r , while the other 18 sign-table cells are free. $\square$

For a relation $A \subseteq \{-1, 0, +1\}^2$, write

$$\text{Source}(A) = \{x : (x, y) \in A \text{ for some } y\}, \quad \text{Target}(A) = \{y : (x, y) \in A \text{ for some } x\}.$$

Lemma 2.2 (Safety and saturation). *The projected family is safe if and only if*

$$\text{Target}(A_r) \cap \text{Source}(A_{r+2}) = \emptyset \quad (r \bmod q).$$

If $Y_r = \text{Target}(A_r)$, replacing every relation by

$$\tilde{A}_r = (\{-1, 0, +1\} \setminus Y_{r-2}) \times Y_r \tag{21}$$

preserves safety and weakly increases (19). Conversely, every subset profile (Y_r) in (21) gives a safe relation family.

Proof. Two positive outputs at phases $r, r+2$ are composable precisely when the right coordinate of an edge in A_r equals the left coordinate of an edge in A_{r+2} , which is (20). Hence every source of A_r lies outside Y_{r-2} , while every target lies in Y_r . Thus $A_r \subseteq \tilde{A}_r$. The weights λ_r are nonnegative, so adding the missing edges cannot lower the score. Finally, any edge of $\tilde{A}_r$ ends in Y_r , and any edge of $\tilde{A}_{r+2}$ begins outside Y_r ; therefore the saturated family is safe. $\square$

Lemma 2.3 (Reflection and absolute value). *Define $F_r^\rho(x, z, y) = F_r(-x, -z, -y)$. Reflection preserves universal safety and*

$$L_q(F^\rho) = -L_q(F).$$

Consequently $\max_{\text{safe}} |L_q| = \max_{\text{safe}} L_q$, with both orientations attained.

Proof. Negation is a bijection of all ternary biwords, so it preserves (3). For $H = zF$,

$$zF^\rho(x, z, y) = -H(-x, -z, -y).$$

Every exact-support sign stratum is invariant under simultaneous negation, and its uniform average therefore changes sign. The RH-394 table law gives the displayed identity. $\square$

3 The tropical optimizer and the self-loop exception

Proposition 3.1 (Exact all- q tropical formula). *Formula (12) holds for every finite q , including the self-loop clocks $q = 1, 2$.*

Proof. By Lemmas 2.1–2.2, a positive optimum is obtained by choosing $Y_r \subseteq \{-1, 0, +1\}$, and its phase contribution is exactly

$$\sum_{x \notin Y_{r-2}} \sum_{y \in Y_r} \lambda_r(x, y) = \mathcal{K}_r(Y_{r-2}, Y_r).$$

Addition by two decomposes the phases into $\gcd(q, 2)$ disjoint cycles. The choices on distinct cycles do not interact, so maximizing their sums separately proves (12). Keeping all subsets of $\{-1, 0, +1\}$ gives eight states, and cyclic closure is the tropical trace. Lemma 2.3 converts the positive optimum to the absolute capacity. $\square$

Proposition 3.2 (Four-state compression for $q \geq 3$). *If $q \geq 3$, an optimizer can be chosen so that every Y_r contains either both nonzero signs or neither. On these four states the transition is (13).*

Proof. The density $\lambda_r(x, y)$ depends only on whether each coordinate is zero. Hold fixed the membership of zero in every Y_r , and put $k_r = |Y_r \cap \{-1, +1\}| \in \{0, 1, 2\}$. When $q \geq 3$, the two occurrences of Y_r are in distinct transition terms: it is the target of the r -term and the excluded source of the $r + 2$ -term. Holding all other variables fixed, the objective is affine in k_r . Replacing an interior value $k_r = 1$ by one of the endpoints 0, 2 therefore does not lower the objective. Repeating this finitely many times yields an antipodally symmetric optimizer.

For such a set write $u_0 = \mathbf{1}_{\{0 \in Y\}}$ and $u_1 = k/2$. Splitting (11) according to the four zero/nonzero support patterns gives the four terms in (13). For $q = 1, 2$, the same variable occurs as both source and target in a self-loop and produces $k_r(2 - k_r)$; the affine argument is therefore unavailable. $\square$

Proposition 3.3 (Small clocks and genuine memory gain). *The values in (14)–(16) hold, and*

$$C(6) - F_{\text{one}}(6) = \frac{2K_2 - K_1}{4} > 0, \quad F_{\text{one}}(6) = \frac{3K_1}{8}.$$

Proof. For $q = 1, 2$, substitute (7)–(10) into $\mathcal{K}_r(Y, Y)$ and check the eight subsets. At $q = 1$, the zero-only state is optimal and gives $K_2 - K_3$. At $q = 2$, the even phase is optimized by a one-sign state, contributing $(K_2 - K_3)/4$, while the odd phase is optimized by the zero-only state, contributing $K_2/2$. Hence

$$C(2) = \frac{K_2 - K_3}{4} + \frac{K_2}{2} = \frac{3K_2 - K_3}{4}.$$

The antipodally symmetric four-state restriction would instead give $K_2 - K_3$, so the $q = 2$ exception is strict.

For $q = 3, 4, 6$, Proposition 3.2 reduces the exact calculation to the following four-state max-plus products. Each row lists the resulting coefficient vector in the basis (K_1, K_2, K_3) :

q	coefficient vector	value
3	$(3/8, 0, 0)$	$3K_1/8$
4	$(2/3, 0, 0)$	$2K_1/3$
6	$(1/8, 1/2, 0)$	$K_1/8 + K_2/2$

These are finite products of four-by-four matrices with entries (13); comparing the four possible successor states at each step gives the displayed maxima. The full eight-state calculation agrees for $q \geq 3$. Explicit maximizing profiles, listed in +2 cycle order, are

q	selector states
3	$(Y_0, Y_2, Y_1) = (\emptyset, \{-1, 0, +1\}, \emptyset),$
4	$(Y_0, Y_2) = (\emptyset, \{-1, 0, +1\}), \quad (Y_1, Y_3) = (\{-1, 0, +1\}, \emptyset),$
6	$(Y_0, Y_2, Y_4) = (\emptyset, \{-1, 0, +1\}, \emptyset), \quad Y_1 = Y_3 = Y_5 = \{0\}.$

Substitution into (13) gives the table entries. The other successor states are dominated using only the comparisons below, so no numerical approximation enters the finite maximum.

For completeness, the only Euler-product comparisons needed to order the finite candidates are

$$\frac{1}{3} < \frac{K_3}{K_2} < \frac{1}{2}, \quad \frac{K_2}{K_1} > \frac{1}{2}. \quad (22)$$

The upper bound in the first inequality follows from the $p = 2$ factor

$$\frac{K_3}{K_2} = \frac{1}{2} \prod_{p \geq 3} \left(1 - \frac{1}{p^2 - 2}\right).$$

For the lower bound, use $\prod(1 - x_i) \geq 1 - \sum x_i$ and

$$\sum_{p \geq 3} \frac{1}{p^2 - 2} < \sum_{\substack{n \geq 3 \\ n \text{ odd}}} \frac{1}{n(n-1)} = 1 - \log 2 < \frac{1}{3}.$$

Likewise

$$\frac{K_2}{K_1} = \frac{2}{3} \prod_{p \geq 3} \left(1 - \frac{1}{p^2 - 1}\right) > \frac{2}{3} \left(1 - \sum_{\substack{n \geq 3 \\ n \text{ odd}}} \frac{1}{n^2 - 1}\right) = \frac{1}{2},$$

where the prime sum is a strict sub-sum and the odd-integer sum is $1/4$. This last inequality proves (3.3). The one-site value follows from the phase-MWIS formula recalled next. $\square$

4 Square-support marginal charge

We isolate the finite combinatorial input inherited from RH-375. Define

$$M(q) = \max_{\substack{I \subseteq \mathbb{Z}/q\mathbb{Z} \\ I \cap (I+2) = \emptyset}} \sum_{r \in I} \Theta_{q,r}(\{C\}). \quad (23)$$

Choosing $A_r = \{-1, 0, +1\}^2$ on I and $A_r = \emptyset$ off I embeds this one-site relation family into the centered model, so

$$M(q) \leq C(q). \quad (24)$$

RH-375, Theorem 2.2, equation (7) (printed/PDF page 2), identifies (23) as the weighted phase-MWIS; only this finite formula, not its ordinary-Cesàro analytic formula, is used here [1].

Let $3 = p_1 < p_2 < \dots$ be the odd primes and set

$$q_y = 4 \prod_{i \leq y} p_i^2, \quad A_y = \prod_{i \leq y} (p_i^2 - 1), \quad q_1 = 36. \quad (25)$$

Let O_y be the RH-375 count of odd positive runs on the odd $+2$ -cycle. Its exact square-clock value and limit are

$$B_y = M(q_y) = \frac{4 + 2O_y/A_y}{\pi^2}, \quad (26)$$

$$B_\infty = \lim_{y \rightarrow \infty} B_y = \frac{4 + 2\mathcal{C}}{\pi^2}, \quad (27)$$

$$\mathcal{C} = \frac{1}{e_1} \sum_{j \in \{1, 3, 5, 7\}} (e_j - 2e_{j+1} + e_{j+2}), \quad e_m = \prod_{p \text{ odd}} \left(1 - \frac{m}{p^2}\right).$$

RH-375, equations (10)–(12), Proposition 4.1, equation (13), and Theorem 5.1, equations (16)–(17) (printed/PDF pages 3–5), prove that B_y is strictly increasing to B_∞ . Its Lemma 3.1 and Proposition 3.2, equations (8)–(9) (page 3), supply the finite density aggregation and period-lift precedents.

Lemma 4.1 (Clock divisibility). *If $q \mid Q$, then $C(q) \leq C(Q)$.*

Proof. Regard every q -phase table as a Q -phase table by literal repetition. It produces exactly the same output word, so universal safety and every fixed-table terminal limit are unchanged. The declared period need not be minimal. Maximizing only after the fixed limits gives the claim. No estimate uniform in q is used. $\square$

Lemma 4.2 (Shared-coordinate marginal identity). *Suppose $q_y \mid Q$ and Q has exactly the same prime divisors as q_y . Every phase r with $\Theta_{Q,r}(\{C\}) > 0$ has one common weight*

$$\delta_Q = \Theta_{Q,r}(\{C\}).$$

For two adjacent positive center phases $r, r+2$, define

$$m_r(t) = \sum_{x \in \{-1,0,+1\}} \lambda_r(x, t), \quad m'_{r+2}(t) = \sum_{z \in \{-1,0,+1\}} \lambda_{r+2}(t, z).$$

Then

$$m_r(t) = m'_{r+2}(t) \quad (t \in \{-1,0,+1\}), \quad \sum_{t \in \{-1,0,+1\}} m_r(t) = \delta_Q. \quad (28)$$

Proof. Every prime dividing Q occurs in Q to exponent at least two. Consequently a supported prime square either forces the shared integer $n+1$ to be nonsquarefree in both descriptions or permits it in both. For primes outside the support, the local squarefree pair conditions for $(n, n+1)$ and $(n+1, n+2)$ are translates of one another. Thus their Euler factors coincide for each of the three exact states $\mu_0(n+1) = t$. Uniform sign splitting on nonzero exact-support strata gives the first identity in (28).

Summing over t , and also over the unused outer coordinate, counts every center-squarefree event with center sign $+1$. Its sign density is one half of δ_Q , while the positive projection changes the score by two. Hence the total λ -mass is δ_Q , proving the second identity. $\square$

Proposition 4.3 (Square-support saturation). *Under the hypotheses of Lemma 4.2,*

$$\boxed{C(Q) = M(Q) = M(q_y) = B_y.} \quad (29)$$

Proof. For a saturated selector profile put

$$W_r = \mathcal{K}_r(Y_{r-2}, Y_r).$$

If the center phase has zero density then $W_r = 0$. For two adjacent positive phases, Lemma 4.2 and the safety split at the shared state t give

$$\begin{aligned} W_r + W_{r+2} &\leq \sum_{t \in Y_r} m_r(t) + \sum_{t \notin Y_r} m'_{r+2}(t) \\ &= \sum_{t \in \{-1,0,+1\}} m_r(t) = \delta_Q. \end{aligned}$$

Also $W_r \leq \delta_Q$. The phases forced to zero modulo 4 and 9 split the two $+2$ -cycles into finite positive runs. Pairing successive sites in a run of length ℓ shows that its total contribution is at most $\lceil \ell/2 \rceil \delta_Q$, exactly the MWIS value of that path. Summing the runs proves $C(Q) \leq M(Q)$. The reverse inequality is the one-site embedding (24).

Write $Q = Rq_y$. The positive support word is R literal repetitions of the q_y word, and $\delta_Q = \delta_{q_y}/R$. Forced 4- and 9-zero phases prevent runs from joining across a repetition boundary. Thus the path-MWIS count scales by R while its weight scales by $1/R$, so $M(Q) = M(q_y) = B_y$. This is the special same-prime-support saturation of RH-375, now reached through the centered marginal charge; it is not a general cyclic-cover law. $\square$

5 All-clock rigidity and strict nonattainment

Proof of the all-clock statement in Theorem 1.1. Fix any finite q . Choose y so that every odd prime divisor of q is among $p_1, \dots, p_y$, and put

$$Q = \text{lcm}(q, q_y).$$

Then $q \mid Q$, $q_y \mid Q$, and Q has exactly the prime support of q_y . Lemma 4.1, Proposition 4.3, and the strict increase of the RH-375 endpoints give

$$C(q) \leq C(Q) = B_y < B_\infty.$$

This proves the all-clock upper bound and strict nonattainment at every finite clock.

Conversely, the embedded one-site relations at the square clocks satisfy

$$C(q_y) \geq M(q_y) = B_y.$$

Letting $y \rightarrow \infty$ gives $\sup_{q < \infty} C(q) \geq B_\infty$, and the preceding upper bound gives the reverse inequality. The order of operations is essential: q and the table family are fixed, then $X \rightarrow \infty$, then the finite safe maximum is formed, and only afterward is the scalar supremum over finite clocks taken. $\square$

Remark 5.1 (Where memory helps and where it does not). The $q = 2$ one-sign self-loop and the strict $q = 6$ gap (3.3) show that the centered relation class is genuinely larger than the one-site class at a fixed clock. Proposition 4.3 shows, by a coordinatewise marginal charge rather than a flat one-site argument, that this gain disappears on the cofinal square-support family. The endpoint rigidity is therefore not a rephrasing of RH-375.

6 Source roles, executable audit, and limitations

The sole terminal-log analytic input is the complete fixed three-shift table law of RH-394 [2]. RH-375 contributes only finite squarefree phase densities, the one-site MWIS, clock lift, square-clock values B_y , same-prime-support combinatorics, and the endpoint B_∞ [1]. In particular, its ordinary-Cesàro formula is not used as a terminal-clock theorem. Tao’s two-point theorem and the odd-correlation theorem of Tao–Teräväinen are inherited provenance inside RH-394, not newly strengthened here [3, 4]. The two frozen repository identities are 6b3d616851cd2d7cba66371d0aa9f25b8e8bf2f7 for RH-394 and 071fed1b2a5d8488b9d2e35a99a753953b233584 for RH-375.

The companion certificate performs exact relation-composition, positive-projection, reflection, phase-density, full eight-state, four-state, self-loop, small-clock, marginal, run-charge, lift, and firewall checks. It uses rigorous rational enclosures for the Euler products and rejects unresolved comparisons. This is a finite reproduction and regression artifact, not a substitute for the analytic table law or for the proofs above.

The immutable source closure has 148 Git objects and four ordered remote logical objects, hence 152 logical objects. The four external roles are, in order, Johnston–Yang closure-only, Maynard closure-only, Tao’s inherited two-point input, and Tao–Teräväinen’s inherited odd-correlation input. Their recorded release-redistribution flags are no, no, yes, and no. All external PDFs remain nonvended, source replay is offline by default, and all six sealed payload identities are excluded from the release tree.

The centered rule (2) explicitly reads $\mu(n + 1)$. It is not the causal window-end model or a physical online controller; the causal specialization is a separate, already bounded route. The theorem keeps q, F, ω fixed before the terminal limit. It gives no ordinary Cesàro limit, effective rate, growing $q = q(X)$, growing table family, maximum before the limit, adaptive finite-prefix capacity, generic graph-coupled capacity, or even odd-support correlation of order at least four. It makes no operator, trace-formula, zero-identification, Riemann-hypothesis, or Gates A–E claim.

Declarations

Data availability. The finite certificate, exact result and schema, source-lock metadata, and offline verification code are included in the accompanying RH-395 package. No new empirical data were generated and no external source PDF is redistributed.

Ethics declaration. This mathematical study involved no human participants, animals, personal data, or sensitive datasets; institutional ethics approval was not applicable.

Author contributions (CRediT). The RH research program performed conceptualization, formal analysis, methodology, software, validation, and writing (original draft, review, and editing).

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